

Adaptive Weighting for Time-to-Event Continual Reassessment Method: Improving Safety in Phase I Dose-Finding Trials

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40.6% reduction in overdosing while maintaining MTD selection accuracy — a drop-in replacement for TITE-CRM.

Background & Motivation

Phase I trials seek the **maximum tolerated dose (MTD)** — the dose with DLT probability closest to target $p^* = 0.25$.

The Problem: Modern oncology agents produce **delayed-onset toxicities** appearing weeks to months later. TITE-CRM handles incomplete follow-up with a fixed linear weight $w = t_i/T_{\max}$ that assumes toxicity risk rises *uniformly* — an ad hoc rule with no probabilistic justification.

When the true delay pattern is non-linear, TITE-CRM escalates too aggressively, **exposing patients to excessive toxicity**.

Our Innovation: Adaptive Predictive Weights

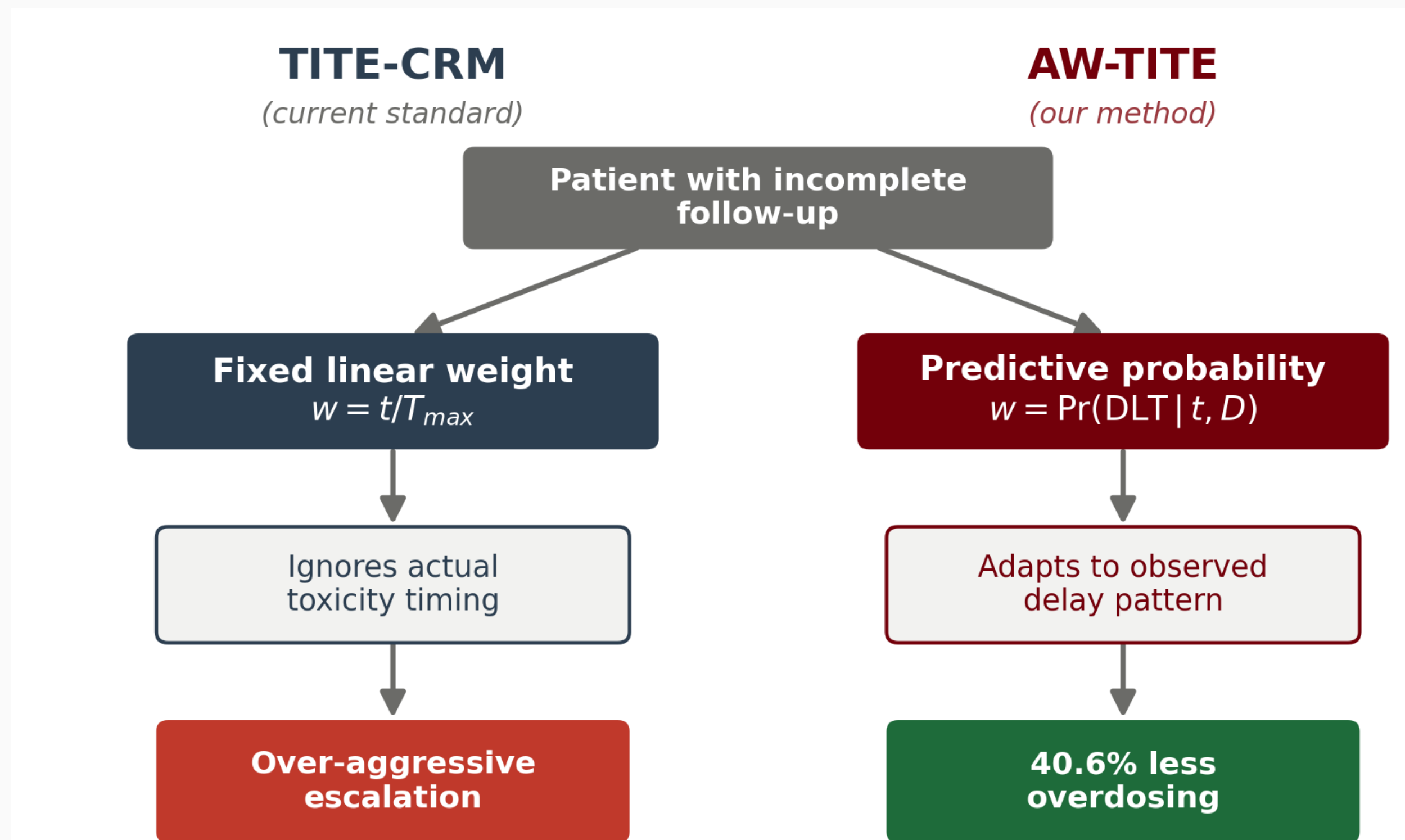
Replace TITE-CRM's ad hoc weight with the **posterior predictive probability** of eventual toxicity, given partial follow-up:

$$w_i = \Pr\left(T \leq T_{\max} \mid T > t_i, \mathcal{D}\right)$$

Under a **Weibull timing model**, this has a simple closed form:

$$w_i = 1 - \exp\{-\hat{\lambda}(d_i) \Delta_i\}, \quad \hat{\lambda}_{\text{MLE}}(d) = \frac{\sum_j \delta_j}{\sum_j u_j^\gamma}$$

- **Data-driven, interpretable weights** that adapt to the *observed* delay pattern — no change to the standard CRM workflow.



Simulation Study Design

- **Trial:** $N = 30$, $K = 5$ doses, $p^* = 0.25$, $T_{\max} = 12$ wks, accrual every 2 weeks
- **DLT times:** Weibull($\gamma = 2.0$) with dose-specific rates
- **Replications:** 2,000 per scenario
- **Methods:** AW-MLE, AW-BAYES, TITE-CRM, 3+3, mTPI, BOIN

Scenario	MTD	True DLT Probabilities
Standard	d3	0.05, 0.10, 0.20, 0.35, 0.50
Steep Curve	d4	0.02, 0.05, 0.10, 0.25, 0.50
Flat Curve	d4	0.10, 0.15, 0.20, 0.25, 0.30

Table 1. Three dose-toxicity scenarios evaluated.

Primary Result

Averaged across all three scenarios, AW-MLE **reduced the fraction of patients above the MTD from 0.340 (TITE-CRM) to 0.202** — a 40.6% relative reduction.

Mean diff.: -0.139 95% CI: $[-0.210, -0.067]$ $p < 0.001$

MTD accuracy was **fully maintained** (mean diff. $+0.023$, $p = 0.21$).

- **Standard:** 33.1% reduction (0.279 vs. 0.417)
- **Steep curve:** 37.4% reduction (0.112 vs. 0.179)
- **Flat curve:** 49.6% reduction (0.213 vs. 0.423) — largest gain where discrimination is hardest

Figure 1: Fraction of Patients Above MTD (Lower = Safer)

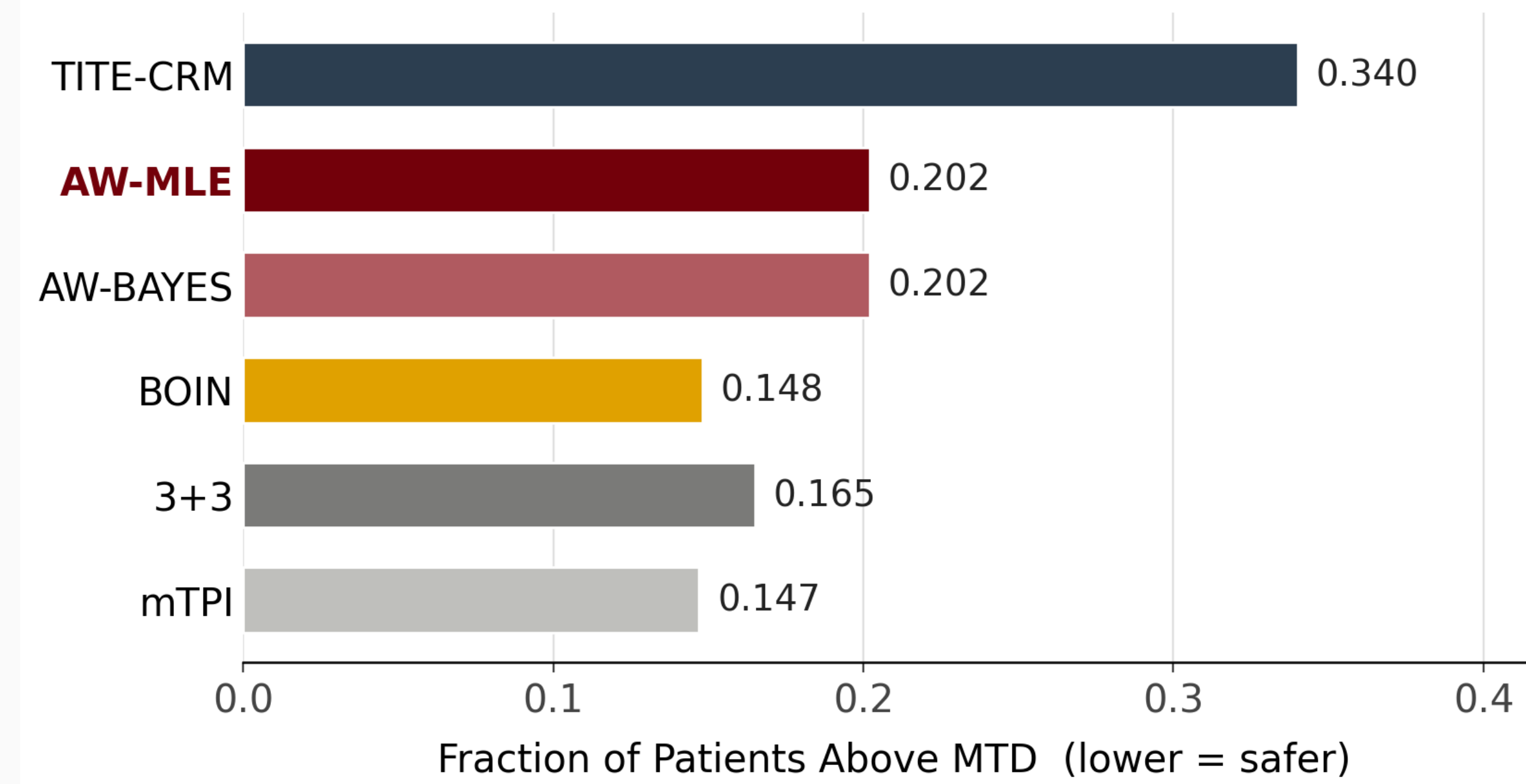


Figure 1. Averaged across Standard, Steep, and Flat scenarios. AW-MLE achieves 40.6% lower overdosing than TITE-CRM.

Figure 2: P(Correct MTD Selection) by Scenario

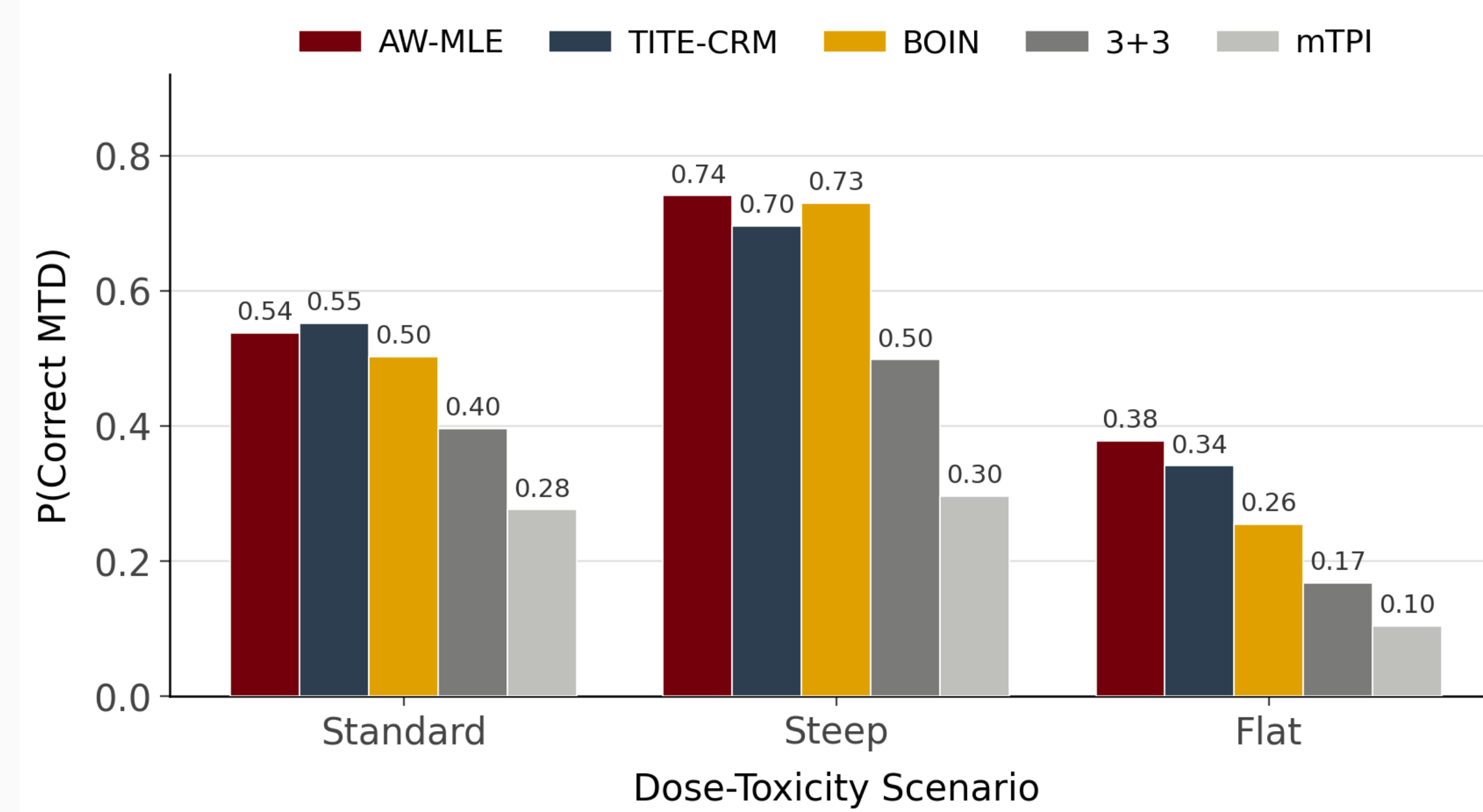


Figure 2. AW-MLE achieves highest or joint-highest accuracy in all scenarios. mTPI struggles in the flat curve scenario (10.4%).

Conclusions

- **40.6% fewer patients overdosed** vs. TITE-CRM ($p < 0.001$) — ~ 4 per 30-patient trial.
- **MTD accuracy fully maintained** ($p = 0.21$). No safety–accuracy tradeoff.
- **Drop-in replacement:** same CRM workflow and safety rules.

Operating Characteristics — All Methods & Scenarios

	Standard		Steep		Flat	
AW-MLE	0.54	0.28	0.74	0.11	0.38	0.21
TITE-CRM	0.55	0.42	0.70	0.18	0.34	0.42
BOIN	0.50	0.23	0.73	0.10	0.26	0.12
3+3	0.40	0.22	0.50	0.14	0.17	0.13
mTPI	0.28	0.20	0.30	0.13	0.10	0.09
	P(Correct) ↑ better	Above MTD ↓ better	P(Correct) ↑ better	Above MTD ↓ better	P(Correct) ↑ better	Above MTD ↓ better

AW-MLE pairs top accuracy (green, left) with low overdosing across all scenarios. TITE-CRM's overdosing column is consistently red.

Sensitivity Analyses

Robust across all analyses (1000 simulations per value):

Parameter	Range	CV	Finding
Accrual rate	1–4 weeks	2.1%	Robust
Sample size N	20–50 patients	—	Consistent ranking
Shape γ	1.5–3.0	2.3%	Negligible effect
$T_{\max}$	8–16 weeks	—	Plateau at 12–14w
Prior (Bayesian)	Weak/Med/Strong	—	Weak $\approx$ MLE

Table 2. CV = coefficient of variation across parameter values.

Key recommendation: Fix $\gamma = 2.0$. Performance penalty for $\pm 50\%$ misspecification is $< 3\%$ variation in accuracy.

How to Implement AW-TITE

1. **Shape parameter:** fix $\gamma = 2.0$ (robust across 1.5–3.0).
2. **Observation window:** $T_{\max} = 12$ weeks.
3. **Weight updates:** compute the adaptive weight w_i (see equation above) for each patient with incomplete follow-up.
4. **Safety rules:** standard CRM — no dose-skipping, start low, ≥ 3 patients before de-escalation.

Comparison vs. Algorithm-Based Methods

Comparison	Mean Diff.	95% CI	p -value
AW-MLE vs. mTPI	+32.6 pp	[26.2, 44.5]	< 0.001
AW-MLE vs. 3+3	+19.8 pp	[14.3, 24.3]	< 0.001
AW-MLE vs. BOIN	+5.6 pp	[1.1, 12.4]	0.018

Table 3. P(Correct MTD) differences vs. algorithm-based designs, averaged across 3 scenarios. pp = percentage points.

In the flat curve scenario, mTPI selects dose 1 in $\approx 60\%$ of trials (severe under-escalation); AW-MLE achieves 37.8% correct selection.

References

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2. O'Quigley, J., Pepe, M., Fisher, L. (1990). Continual reassessment method. *Biometrics* 46:33–48.
3. Yuan, Y., Yin, G. (2011). Outcome-adaptive randomization. *JCO* 29:e390–e392.